

Computer Science & IT



Compiler Design



Parsers
Lecture No. 03



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Topics to be Covered



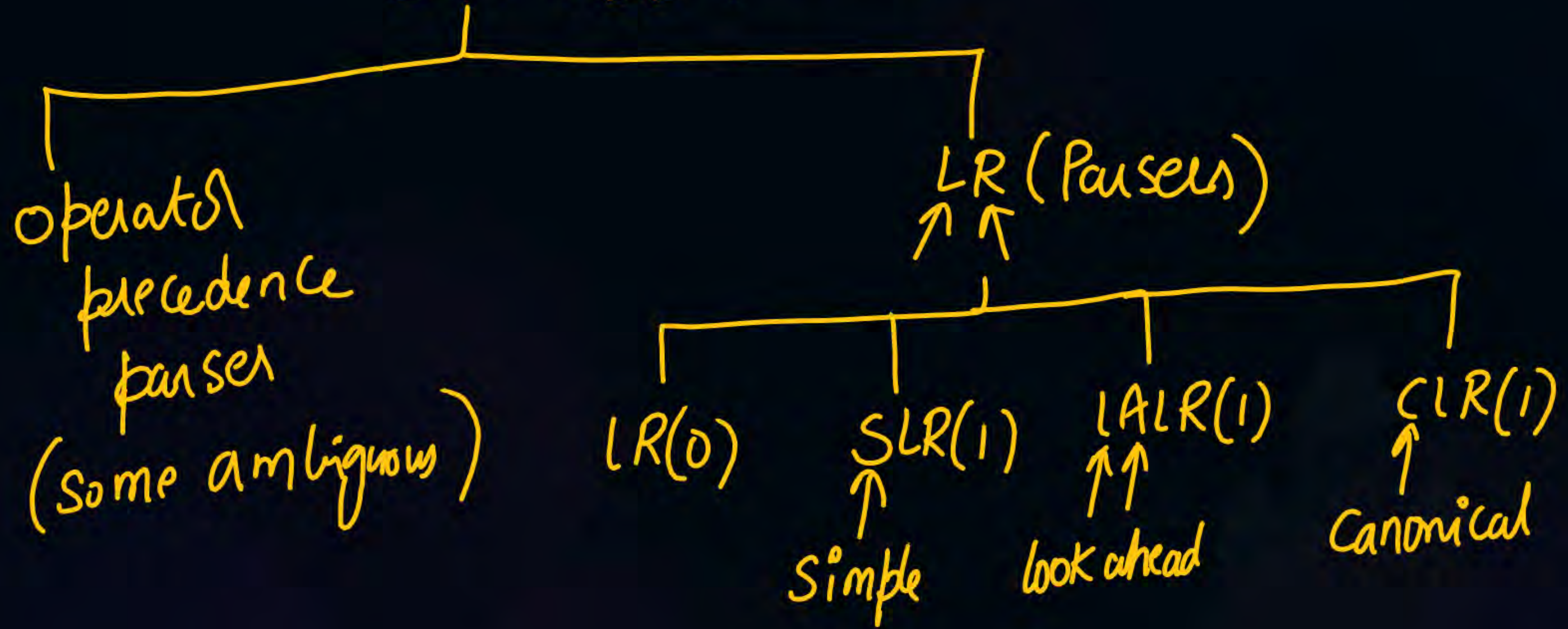
Topic

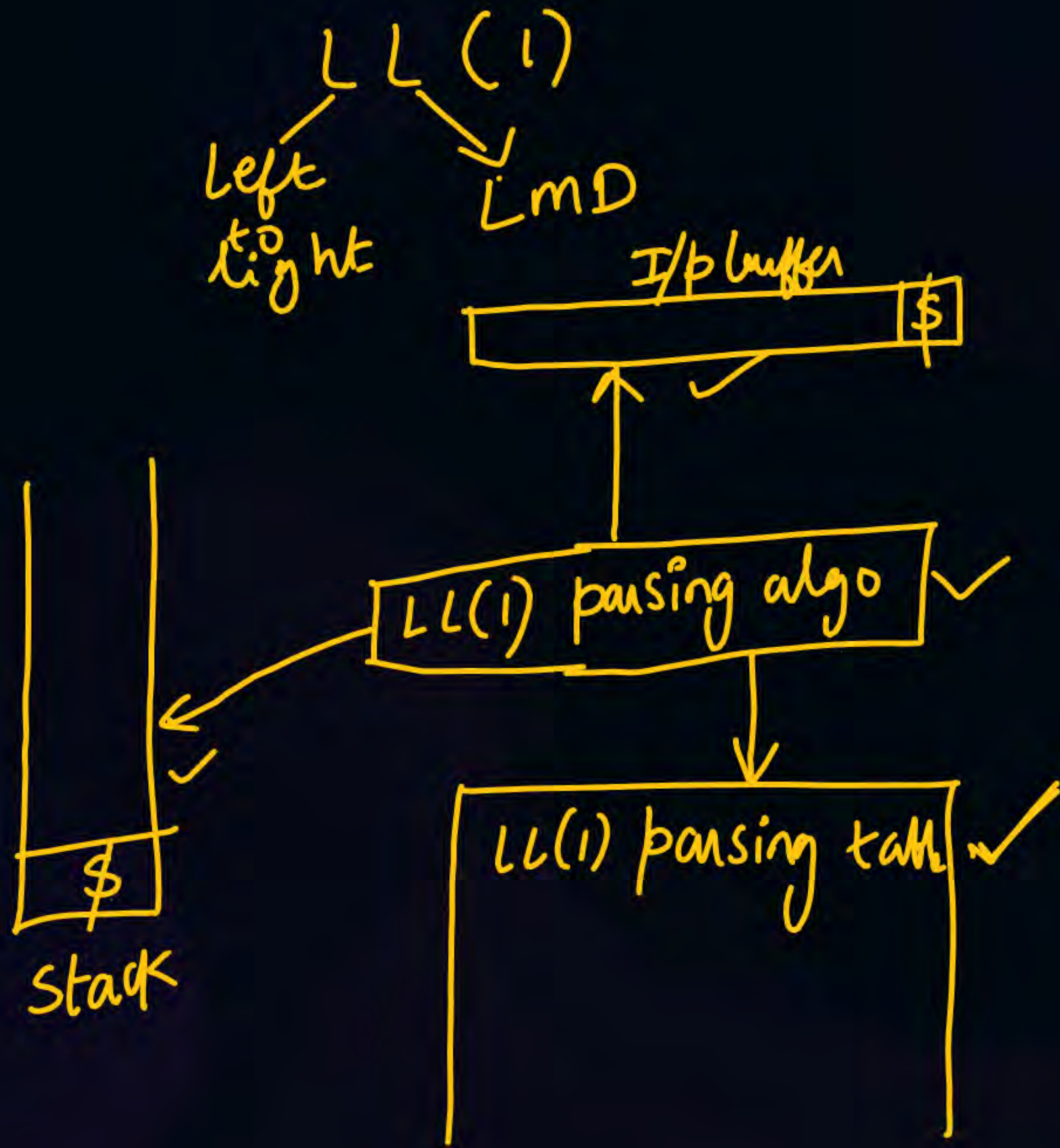
Panser

Topic

Topic

BUP (SR parsers)





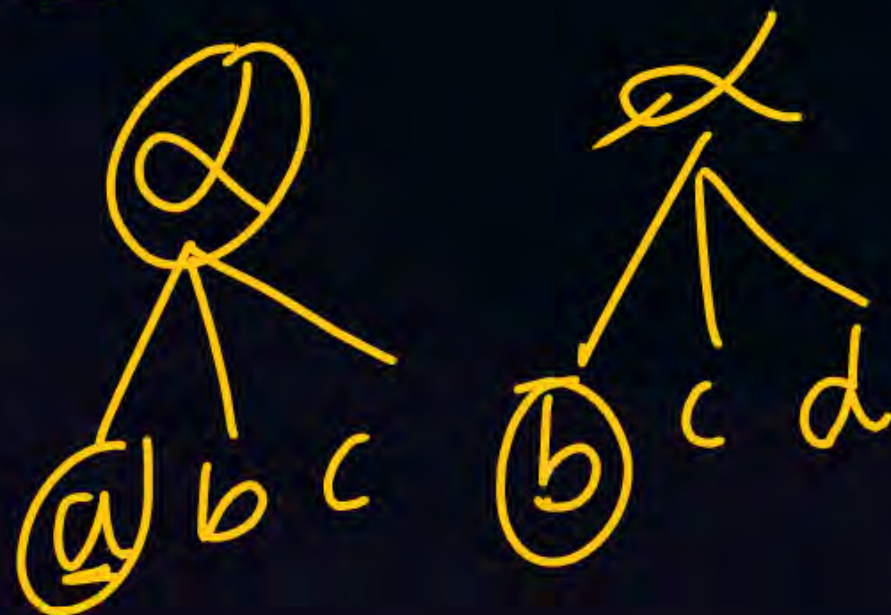
LL(1) parsing parsing algo will take i/p from
 i/p buffer, top of the stack from stack
 and consults LL(1) parsing table to take a
 decision

We need to first construct the parsing table
using First and Follow .

First(α) , where α is any string of grammar symbols:

First of α gives set of all terminals that begin the
Strings derived from ' α '.

First(α) Follow(A)



✓
index funds

5K

10K

20K

1K

If $\alpha \xRightarrow{*} \epsilon$ and $\alpha \Rightarrow \epsilon$, then ϵ is in $\text{First}(\alpha)$

Rules to find $\text{First}(\alpha)$:-

If α is a terminal, then $\text{First}(\alpha) = \alpha$

Ex:- $\text{First}(+) = \{+\}$ $\text{First}(*) = \{*\}$ $\text{First}(\text{id}) = \{\text{id}\}$

If α is a non terminal then there are two possibilities

a) $\alpha \rightarrow \epsilon$ then add ' ϵ ' to $\text{First}(\alpha)$

$\alpha \xRightarrow{*} \epsilon$, then add ' ϵ ' to $\text{First}(\alpha)$

b) $\alpha \rightarrow \gamma_1 \gamma_2 \gamma_3 \dots \gamma_k$ if $\gamma_1 \xRightarrow{*} \epsilon$
 then $\text{First}(\alpha) = \text{First}(\gamma_1)$ if $\gamma_1 \xRightarrow{*} \epsilon$
 $\text{First}(\alpha) = \text{First}(\gamma_1) - \{\epsilon\} \cup \text{First}(\gamma_2 \gamma_3 \dots \gamma_k)$ if $\gamma_1 \xRightarrow{*} \epsilon$

$S \rightarrow \underline{a}ABCD$

$A \rightarrow \underline{a}/\underline{\epsilon}$

$B \rightarrow \underline{b}/\underline{\epsilon}$

$C \rightarrow \underline{d}$

$D \rightarrow \underline{\epsilon}$

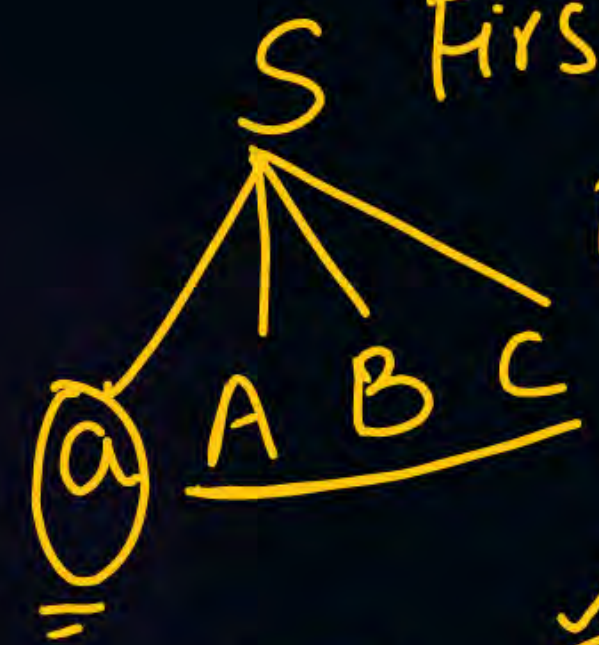
$\text{First}(S) = a$

$\text{First}(A) = \{a, \epsilon\}$

$\text{First}(B) = \{b, \epsilon\}$

$\text{First}(C) = \{d\}$

$\text{First}(D) = \{\epsilon\}$



2014

CS ✓

$$\underline{S} \rightarrow \cancel{A} \cancel{B} \underline{C} \underline{D} \{a, b, d\}$$

$$A \rightarrow \underline{a} / \underline{\epsilon} \quad \{\underline{a}, \underline{\epsilon}\}$$

$$B \rightarrow \underline{b} / \underline{\epsilon} \quad \{\underline{b}, \underline{\epsilon}\}$$

$$C \rightarrow \underline{d} \quad \{d\}$$

$$D \rightarrow \underline{\epsilon} \quad \{\epsilon\}$$

S
\$

(S)\$

$S \rightarrow aA$

aA\$

(sentential form)\$

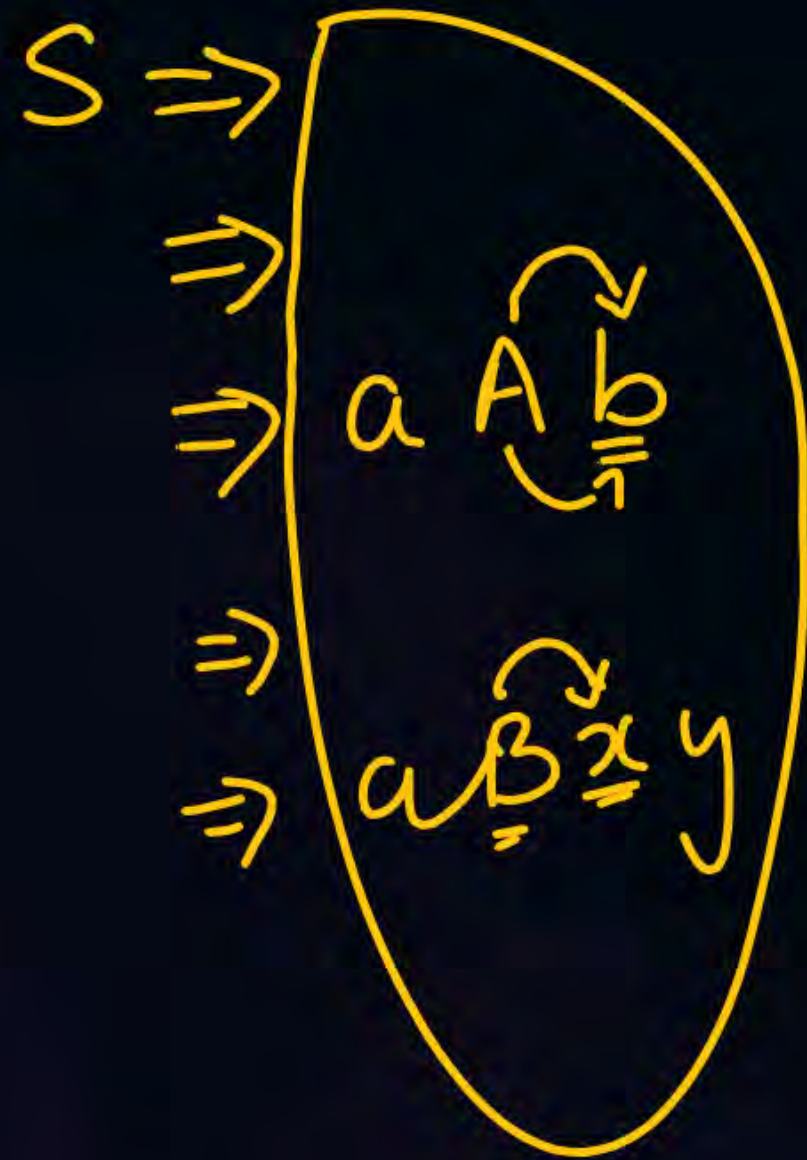
S\$✓

aA\$✓

aAB\$✓

aACD\$✓

Follow $\rightarrow \in X$



RHS of production

$$\underline{A} \rightarrow \underline{B} \neq$$

$$A \rightarrow B \neq^e$$

$$\text{Follow}(B) = \text{First}(\alpha)$$

$$\underline{A} \rightarrow \underline{B}$$

Aa
 Ba

$$\underline{A} \rightarrow \underline{a.b.B}$$

Aa
 $abBa$

Follow(A), where 'A' is a non terminal :-

Follow(A) gives set of all terminals that appear immediately to the right of 'A' in some Sentential form

→ If 'A' is a start symbol and \$ is the i/p right end marker, the place \$ in follow(A)

→ If there is a production $\underline{B} \rightarrow \underline{aA\beta}$, where β is a string of grammar symbols, the first(β) except 'ε' is placed in $\text{follow}(A)$

→ If there is a production $\underline{B} \rightarrow \underline{aA} \Delta$ and $B \rightarrow aA\underline{\beta}$ and β contains 'ε', then everything in $\text{follow}(B)$ is in $\text{follow}(A)$

→ In order to find follow, see the RHS of productions



	first	follow
$S \rightarrow \underline{A} \underline{B} \underline{C} \underline{D}$	$S = \{b, c\}$	$\{\$ \}$
$\underline{A} \rightarrow \underline{b} / \underline{e}$	$\{\underline{b}, \underline{e}\}$	$\{c\}$
$\underline{B} \rightarrow \underline{c}$	$\{c\}$	$\{d\}$
$\underline{C} \rightarrow \underline{d}$	$\{d\}$	$\{e\}$
$\underline{D} \rightarrow \underline{e}$	$\{e\}$	$\{\$ \}$

$S \rightarrow$

$S \rightarrow \underline{A} \underline{B} \underline{C} \underline{D}$

$Foll(\underline{A}) = First(\underline{B} \underline{C} \underline{D})$

$\underline{S} \underline{\$}$
 $\underline{A} \underline{B} \underline{C} \underline{D} \underline{\$}$

$\underline{S} \underline{\$}$ ✓
 $\underline{A} \underline{B} \underline{C} \underline{D} \underline{\$}$

follow $\rightarrow$ RHS First $\rightarrow \epsilon$
 follow $\rightarrow \epsilon$

	first	follow
$S \rightarrow \underline{A} B C D E$	$\{a, b, c\}$	$\{\$ \}$
$A \rightarrow a / \epsilon$	$\{a, \epsilon\}$	$\{b, c\}$
$B \rightarrow b / \epsilon$	$\{b, \epsilon\}$	$\{c\}$
$C \rightarrow c$	$\{c\}$	$\{d, e, \$ \}$
$D \rightarrow d / \epsilon$	$\{d, \epsilon\}$	$\{e, \$ \}$
$E \rightarrow \underline{e} / \epsilon$	$\{e, \epsilon\}$	$\{\$ \}$

(CS) ✓

$\text{Follow}(D) = \text{First}(\overline{E})$
 $\text{Follow}(C) = \text{First}(\overline{D E})$
 $\text{Follow}(A) = \text{First}(\overline{B C D E})$
 $\text{Follow}(B) = \text{First}(\overline{C D E})$

	first	follow.
$S \rightarrow \underline{B} \underline{b} / \underline{C} \underline{d}$	$\{a, b, c, d\}$	$\{\$ \}$
$B \rightarrow \underline{a} \underline{B} / \underline{\epsilon}$	$\{a, \epsilon\}$	$\{b\} \checkmark$
$C \rightarrow \underline{c} \underline{C} / \underline{\epsilon}$	$\{c, \epsilon\}$	$\{d\}$

$$\text{First}(S) = \text{First}(\underline{B} \underline{b}) \cup \text{First}(\underline{C} \underline{d})$$

	First	Follow
$S \rightarrow \cancel{A} \cancel{C} \cancel{B} / \cancel{C} \cancel{b} B / \cancel{B} \underline{a}$ $A \rightarrow \underline{a} a / \cancel{B} \cancel{\epsilon}$	$\{d, g, h, \epsilon, b, a\}$ $\{d, g, h, \epsilon\}$	$\{\$ \}$ $\{h, g, \$ \}$
$B \rightarrow \underline{g} / \epsilon$ $C \rightarrow \underline{h} / \epsilon$	$\{g, \epsilon\}$ $\{h, \epsilon\}$	$\{\$, a, h, g\}$ $\{g, \$, b, h\}$

$S \rightarrow ACB / cbB / \underline{B} \underline{a}$
 $A \rightarrow \underline{B} / \epsilon$

$S \rightarrow A \cancel{C} \cancel{B}$
 $Follow(A) = First(\cancel{C} / \cancel{B})$
 $S \rightarrow AC \cancel{B} / \underline{C} \underline{b} \underline{B}$
 $A \rightarrow \underline{B} \underline{C}$



	first	follow
$S \rightarrow \underline{a} A \underline{B} b$	$\{a\}$	$\{\$, b\}$
$A \rightarrow c/\epsilon$	$\{c, \epsilon\}$	$\{d, b\}$
$B \rightarrow d/\epsilon$	$\{d, \epsilon\}$	$\{b\}$

	first	follow
$S \rightarrow a B \phi h$	$\{a\}$	$\{\$ \}$
$B \rightarrow c C$	$\{c\}$	$\{g, f, h\}$
$C \rightarrow b C / \epsilon$	$\{b, \epsilon\}$	$\{g, f, h\}$
$D \rightarrow \cancel{f} \epsilon$	$\{g, f, \epsilon\}$	$\{h\}$
$E \rightarrow g / \epsilon$	$\{g, \epsilon\}$	$\{f, h\}$
$F \rightarrow f / \epsilon$	$\{f, \epsilon\}$	$\{h\}$

Construction of LL(1) parsing table -

The rows in the table are given by non terminals and columns in the table are given by terminals

	a	b	\$
A	$A \rightarrow \epsilon$		
B			
C			

Algorithm:- For building LL(1) parsing table

- For each production $A \rightarrow \alpha$ of the grammar, perform steps 2 and 3.
- For each terminal 'a' in $\text{First}(\alpha)$, add $A \rightarrow \alpha$ to $m[A, a]$
- If ϵ is in $\text{First}(\alpha)$ add $A \rightarrow \alpha$ to $m[S, b]$ for each terminal 'b' in $\text{follow}(A)$

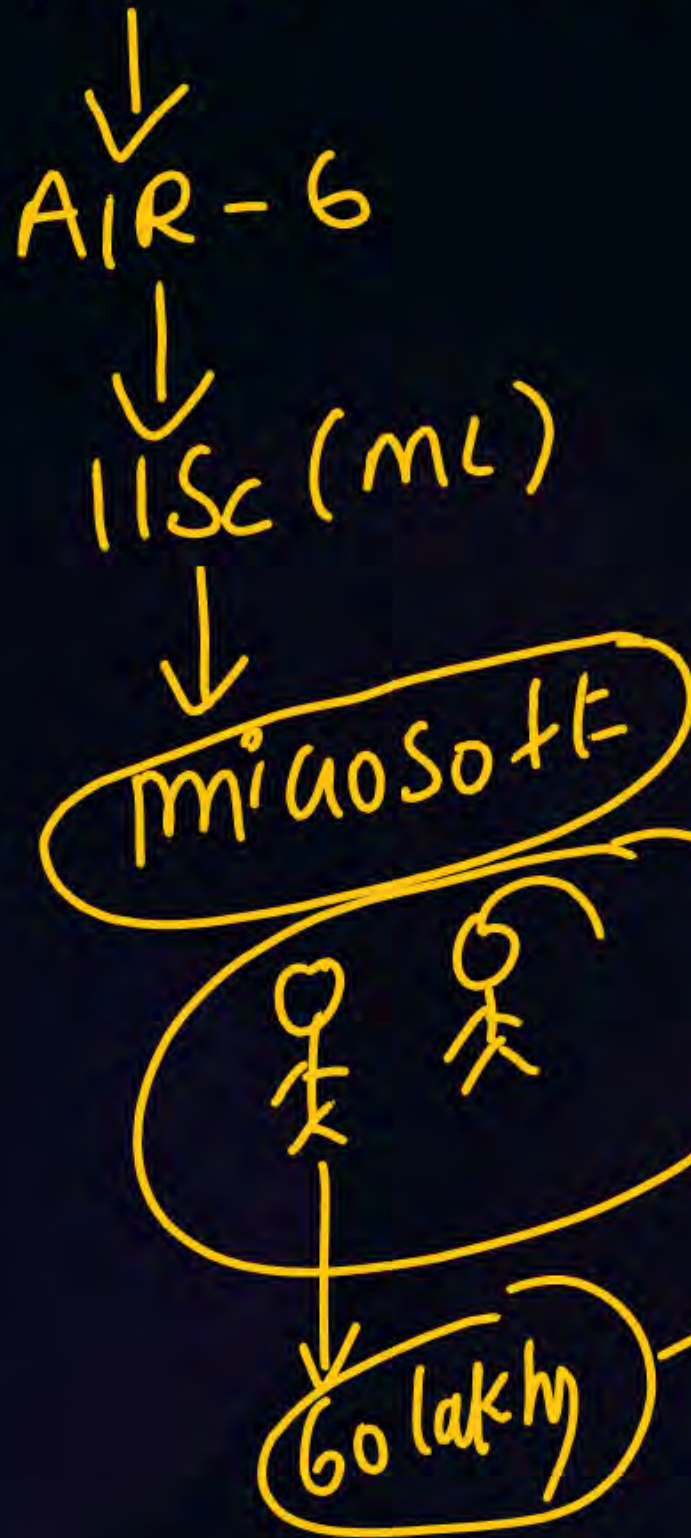
Adity Jain → Student
2019

↓
IITB

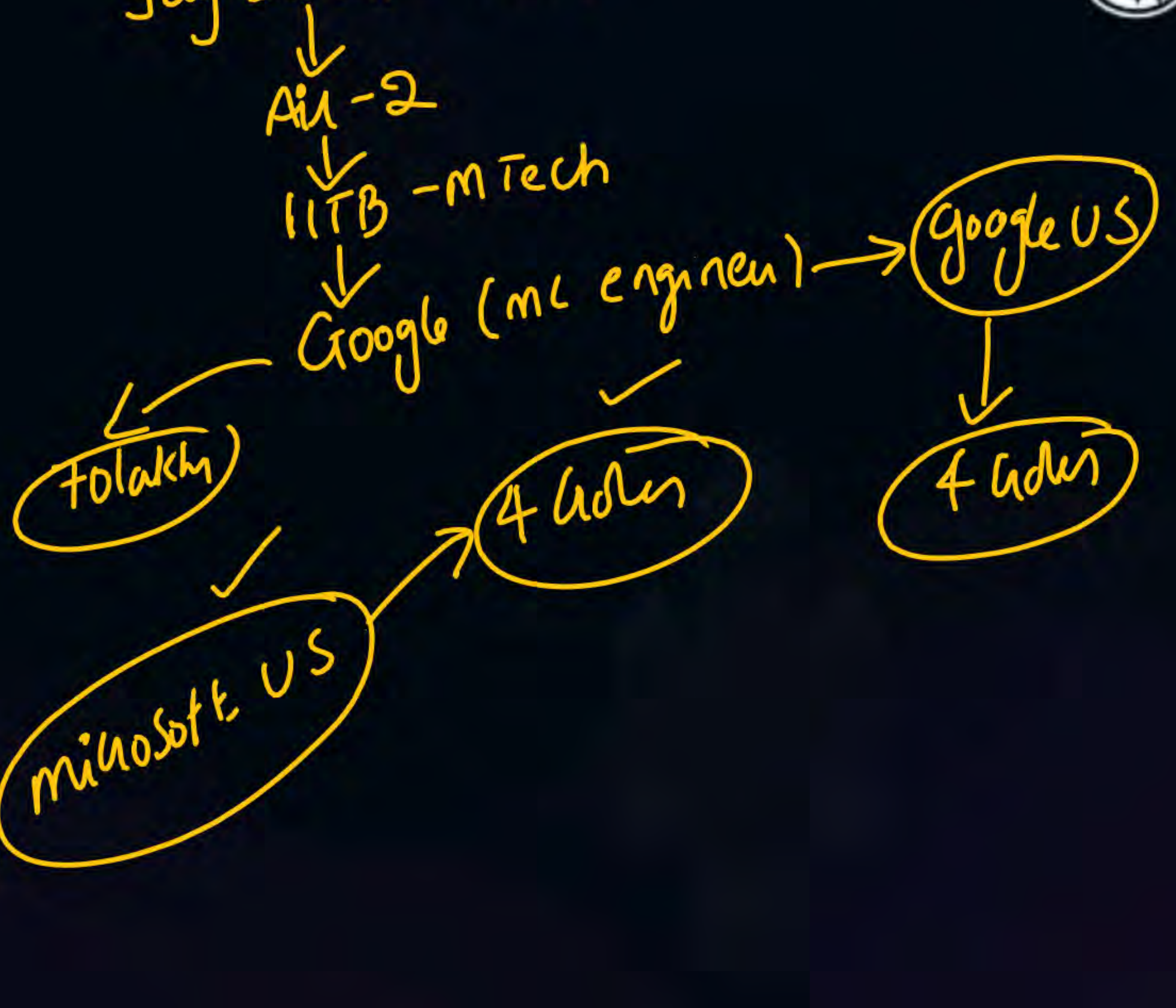
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S/W

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Teaching

Hari - 2019



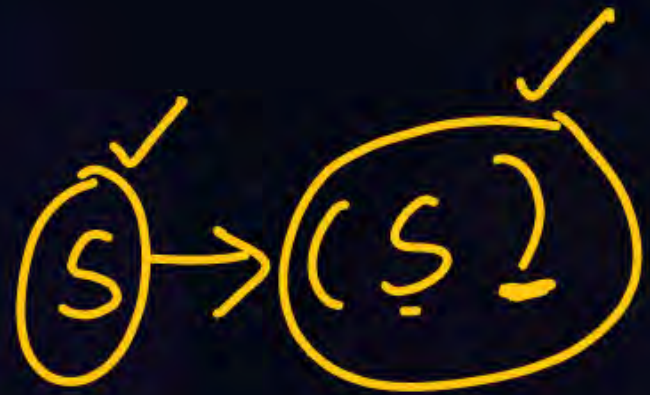
Jay Bansal - 2019



$$S \rightarrow (S) / \epsilon$$

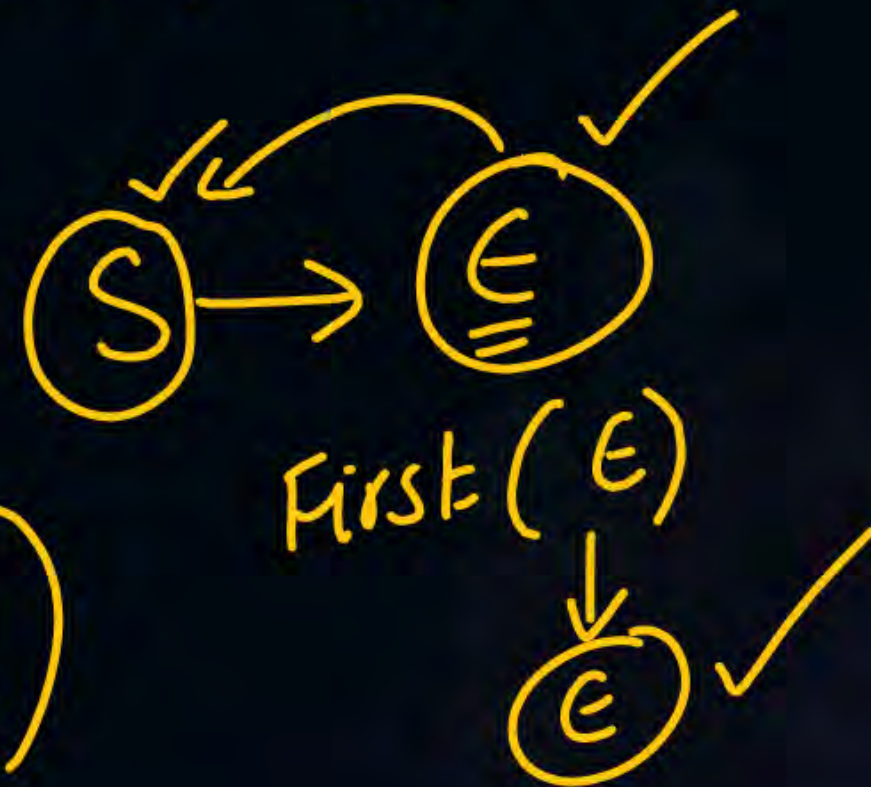
11 (i) parsing table

	(	)	\$
<u>S</u>	$S \rightarrow (S)$	$S \rightarrow \epsilon$	$S \rightarrow \epsilon$



First ((S))

$$\text{Follow}(S) = (\$,)$$



$$S \rightarrow (S) / \epsilon$$

LL(1) parsing table

	(	)	\$
<u>S</u>	<u>$S \rightarrow (S)$</u>	$S \rightarrow \epsilon$	$S \rightarrow \epsilon$

accepted

ϵ
S
$)$
ϵ
S
$)$
S
$\$$

(()) \$
↑ ↑ ↑ ↑ ↑

LL(1)



$$E \rightarrow TE'$$

$$E' \rightarrow +TE' / \epsilon$$

$$T \rightarrow *FT' / \epsilon$$

$$F \rightarrow id / (E)$$



Construct LL(1) parsing table



THANK - YOU